Dependencies and Setup Instructions

Predicting Diabetes Risk from Behavioural and Demographic Health Indicators

EE-439 — Introduction to Machine Learning

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1. Python version

Python 3.14.6 (CPython). Any version from 3.10 upwards should work, but the results in the report were produced on 3.14.6 with the exact package versions below.

2. Packages

Package	Version	Used for
`numpy`	2.5.1	Numerical arrays; storage format for the processed data
`pandas`	3.0.3	Loading the CSV, exploration, and all result tables
`scipy`	1.18.0	Statistical routines used internally by scikit-learn
`scikit-learn`	1.9.0	Preprocessing, the train/test split, three of the four models, cross-validation, permutation importance, and all evaluation metrics
`lightgbm`	4.7.0	The gradient-boosted ensemble; the selected model
`matplotlib`	3.11.1	All 13 figures
`seaborn`	0.13.2	Statistical plots (heatmaps, distributions)
`joblib`	1.5.3	Saving and loading trained models
`ipykernel`	7.3.0	Runs the notebooks
`kagglehub`	latest	Optional. Only if you want notebook 01 to download the dataset itself; the CSV is already in 03_Dataset.

3. Setting it up

Option A — pip and a virtual environment

```
python -m venv .venv

# Windows
.venv\Scripts\activate
# macOS / Linux
source .venv/bin/activate

pip install -r requirements.txt
```

Option B — Conda

```
conda env create -f environment.yml
conda activate brfss-diabetes
```

4. Running the notebooks

Open 02_Code in VS Code, select the environment you created as the kernel, and run the notebooks in numerical order. Or from a terminal:

```
cd 02_Code
jupyter notebook
```

Approximate run times on a 28-core machine:

Notebook	**Time**
`01_data_exploration_and_preprocessing.ipynb`	about 1 minute
`02_model_training_and_selection.ipynb`	about 30 seconds
`03_final_evaluation.ipynb`	about 30 seconds

5. Things that may catch you out

- **File paths.** The notebooks read and write in the current working directory. Copy the CSV from 03_Dataset into 02_Code before running notebook 01, or edit the path in its loading cell.
- **Run them in order.** Notebook 02 needs the .npz that notebook 01 writes, and notebook 03 needs the results file notebook 02 writes. Out of order raises FileNotFoundError.
- **LightGBM installs separately from scikit-learn.** It is not part of scikit-learn and is absent from a default Python install. If the import fails: pip install lightgbm
- **Google Colab.** The notebooks were written for a local environment. To run them in Colab, upload the CSV and change the loading path; everything else works unchanged, since all the libraries used are pre-installed there.
- **You do not need to run anything.** All three notebooks are saved with outputs intact, so every figure and table can be read directly without executing a cell.